

Artemisinin and Chinese medicine as *Tu* science

Jia-Chen Fu

Emory University, United States

The story of discovery of artemisinin highlights the diversity of scientific values across time and space. Resituating artemisinin research within a broader temporal framework allows us to understand how Chinese drugs like *qinghao* came to articulate a space for scientific experimentation and innovation through its embodiment of alternating clusters of meanings associated with *tu* and *yang* within scientific discourse. *Tu* science, which was associated with terms like native, Chinese, local, rustic, mass, and crude, articulated a radical vision of science in the service of socialist revolutionary ideals. *Yang* science, which signified foreign, Western, elite, and professional, tended to bear the hallmarks of professionalism, transnational networks in education and training, and an emphasis on basic or foundational research. With respect to medical research, the case of artemisinin highlights how the constitution of socialist science as an interplay of *tu* and *yang* engendered different scientific values and parameters for scientific endeavor. Modern medical research in Maoist China could harness the productive energies of mass participation to technical expertise in its investigations of Chinese drugs, and under the banner of *tu* science, it became possible and scientifically legitimate to research Chinese drugs in ways that had previously provoked resistance and controversy.

Introduction
The discovery of artemisinin has been presented as a tale resplendent with ingenuity in the face of adversity, social commitment to the good of humanity, genuine esteem for past wisdom, and a heartfelt belief in the value of science. Known in Chinese as *qinghaosu* (青蒿素) and derived from sweet wormwood (*Artemisia annua* L.), artemisinin is the active chemical substance extracted from the Chinese herb *qinghao* (青蒿). That it should be discovered by a young female medical chemist against the backdrop of the Cultural Revolution (1966–1976) and the Vietnam War, as part of a military-supported anti-malarial campaign called Project 523, only accentuates the sense of wonder accompanying the discovery. According to Tu Youyou 屠呦呦 (1930–), who received a Nobel Prize for the discovery

Corresponding author: Fu, J.-C. (jia-chen.fu@emory.edu).
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of artemisinin in 2015 and has become the face associated with the achievement, it happened as follows:

On the basis of collection and analysis of traditional prescriptions, my research group screened over two hundred herbs and three hundred and eighty extracts from them using malarial models of mouse or monkey. I was enlightened by the description, “a handful of *qinghao* immersed with 2 liters of water, get juice and drink it” (Ge Hong, 1956). The antimalarial effect of *qinghao* was gradually cleared up when temperature, enzymolysis, solvents, species, portions, and collecting season of the herb were systematically considered. A new antimalarial was developed in 1971, based on scientific analysis of antimalarial nature of Qinghao with its history of over one thousand years. The new drug won the national award of invention, and brings benefit to the people of the world.¹

Tu’s description has become the standard account for the discovery of artemisinin. Her receipt of both the Lasker–DeBakey Award for Clinical Medical Research in 2011 and the Nobel Prize in Medicine in 2015 has reaffirmed this account, within which several details have been repeated with varying degrees of flourish in all subsequent accounts: the longevity and integrity of traditional Chinese medicine; the integration of traditional medical knowledge with that of biomedicine; and the unassuming power of a single insight in the production of scientific knowledge. As a story we can now tell about the advancement of science, Tu Youyou’s discovery of artemisinin has been molded into the shape of any other story within the medical canon—stories like John Snow’s identification of the Broad Street pump as the source of an 1854 cholera outbreak or Ignaz Semmelweis’s observation that washing one’s hands with anti-septic after working with cadavers and before delivering a child significantly reduced the likelihood of maternal death.

But what does the story of Tu’s discovery tell us about Chinese science in both its socialist and reform era iterations? It may be surprising for an English-speaking

¹ Tu Youyou, “The Development of the Antimalarial Drugs with New Type of Chemical Structure – Qinghaosu and Dihydroqinghaosu,” *Southeast Asian Journal of Tropical Medicine and Public Health* 35, no. 2 (June 2004): 250–51, on 250.

青蒿素与作为“土科学”的中医药

傅家辰

Emory University, United States

青蒿素的发现故事凸显了科学价值观在不同时空下的多样性。将青蒿素研究置于更广阔的时间框架内，使我们能够理解像青蒿这样的中药如何通过其在科学话语中对“土”和“洋”交替意义群的体现，为科学实验和创新开辟了空间。“土科学”与“本土”、“中国”、“地方”、“乡土”、“大众”和“粗糙”等词语相关联，阐明了一种服务于社会主义革命理想的激进科学愿景。“洋科学”则意味着“外国”、“西方”、“精英”和“专业”，往往带有专业性、教育和培训方面的跨国网络以及对基础或奠基性研究的强调等特点。就医学研究而言，青蒿素的案例凸显了社会主义科学作为“土”与“洋”相互作用的构成，如何催生了不同的科学价值观和科学努力的参数。毛泽东时代的中国现代医学研究能够利用大众参与的生产力与技术专长相结合，对中药进行研究，在“土科学”的旗帜下，以过去曾引起抵制和争议的方式研究中药变得可能且具有科学性。

屠呦呦的描述已成为青蒿素发现的标准叙述。她于2011年获得拉斯克临床医学研究奖，并于2015年获得诺贝尔生理学或医学奖，这进一步证实了这一叙述。在所有后续的叙述中，一些细节被反复提及，并带有不同程度的修饰：中医药的悠久历史和完整性；传统医学知识与生物医学的融合；以及单一见解在科学知识生产中不显眼的力量。作为一个我们现在可以讲述的关于科学进步的故事，屠呦呦发现青蒿素的故事已被塑造成医学经典中任何其他故事的形状——例如约翰·斯诺将布罗德街水泵确定为1854年霍乱爆发的源头，或者伊格纳兹·塞梅尔维斯观察到在接触尸体后和分娩前用消毒剂洗手能显著降低产妇死亡的可能性。

引言
青蒿素和中医作为“本土科学”
青蒿素的发现被描绘成一个充满智慧的故事，它面对逆境，致力于人类福祉，真诚地尊重过去的智慧，并由衷地相信科学的价值。青蒿素，中文名为青蒿素（），来源于甜艾（*Artemisia annua* L.），是从中草药青蒿（）中提取的活性化学物质。它是由一位年轻的女医药化学家在文化大革命（1966-1976）和越南战争背景下，作为一项名为“523项目”的军事支持抗疟疾运动的一部分而被发现的，这更增添了发现的奇迹感。根据屠呦呦（1930-）的说法，她因这项发现获得了诺贝尔奖

屠呦呦

通讯作者：傅嘉辰 (jia-chen.fu@emory.edu)。关键词：中医药；生物医学；疟疾；青蒿素；文化大革命；科学价值观；医学研究；中国。在线发布日期：xxxxxx

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2015년 노벨 생리의학상 수상자로 선정되어 이 업적과 관련된 인물이 되었는데, 그 과정은 다음과 같습니다.

在收集和分析传统处方的基础上，我的研究小组使用小鼠或猴子的疟疾模型筛选了两百多种草药和三百八十种提取物。我受到了以下描述的启发：“一把青蒿，二升水浸渍，绞汁饮用”（葛洪，1956）。当系统地考虑青蒿的温度、酶解、溶剂、种类、部位和采摘季节时，青蒿的抗疟作用逐渐明朗。1971年，在对青蒿具有千年历史的抗疟性质进行科学分析的基础上，开发出了一种新型抗疟药。这种新药获得了国家发明奖，并造福了世界人民。¹

屠呦呦的描述已成为青蒿素发现的标准叙述。她于2011年获得拉斯克临床医学研究奖，并于2015年获得诺贝尔生理学或医学奖，这进一步证实了这一叙述。在所有后续的叙述中，一些细节被反复提及，并带有不同程度的修饰：中医药的悠久历史和完整性；传统医学知识与生物医学的融合；以及单一见解在科学知识生产中不显眼的力量。作为一个我们现在可以讲述的关于科学进步的故事，屠呦呦发现青蒿素的故事已被塑造成医学经典中任何其他故事的形状——例如约翰·斯诺将布罗德街水泵确定为1854年霍乱爆发的源头，或者伊格纳兹·塞梅尔维斯观察到在接触尸体后和分娩前用消毒剂洗手能显著降低产妇死亡的可能性。

但是，屠呦呦的发现故事告诉我们，中国科学在其社会主义和改革开放时期分别呈现出怎样的面貌？对于英语使用者来说，这可能令人惊讶。

¹ 屠呦呦，“新型化学结构抗疟药——青蒿素和双氢青蒿素的研发”，《东南亚热带医学与公共卫生杂志》35卷2期（2004年6月）：250–51，其中250页。

audience to learn, but Chinese acclaim for Tu Youyou and her achievements has been stinting and riddled by expressions of dissatisfaction and discontent, even after she was awarded the Nobel Prize in Medicine in 2015. International acclaim of Tu Youyou sparked domestic concern and criticism that she alone should receive such celebration.² Her receipt of the State Preeminent Science and Technology Award, the highest scientific honor in China, on January 9, 2017, can be seen as official acknowledgement of her accomplishments, but it has not dampened the expression of diverging opinions. How, it has been asked, can a “three no’s” (*san wu* 三無) scientist—that is someone with no postgraduate degree, no overseas training, and no membership to the Chinese Academy of the Sciences—be the representative face of Chinese medical achievement?³ Some have challenged her role as *the* discoverer.⁴ Even the story of her discovery has encountered strident criticism for its perceived overemphasis of individual merit and failure to recognize the contributions of the other scientists involved.⁵ This apparent consternation and self-reflection suggests that reform era iterations of Chinese science—however superficially they may resemble global science—retain points of idiosyncratic particularism that we need to take seriously.

This essay explores how the discovery of artemisinin illustrates the diversity and malleability of scientific values across time and space. Resituating artemisinin research within a broader temporal framework allows us to see how Chinese drugs like *qinghao* came to articulate a space for scientific experimentation and innovation through its embodiment of alternating clusters of meanings associated with *tu* and *yang* within scientific discourse. Mao-era scientific and political discourse regularly invoked the importance of *tu* (土) over *yang* (洋). The former, which was associated with terms like native, Chinese, local, rustic, mass, crude, and others, articulated a radical vision of science in the service of socialist revolutionary ideals. The broad masses were to play a formative role in the production of scientific knowledge, and through *tu* science, Mao-era China sought to actualize its commitments to self-reliance,

mass mobilization, and practical application.⁶ *Yang*, in contrast, signified foreign, Western, elite, professional, and similar qualities, and *yang* science tended to bear the hallmarks of professionalism, transnational networks in education and training, and an emphasis on basic or foundational research.

Because the scientific work involved in Maoist drug research straddled multiple realms: laboratory-based medical research, textual analyses of traditional medica books and collectanea, and popular mobilization in the name of public health, re-examining the history of artemisinin research as not just a component of modern Chinese medical history, but as an integral part of the history of modern Chinese science affords us a unique perch from which to consider the relationship between Chinese STM (science, technology, and medicine) and the Cultural Revolution. Indeed, a closer exploration of *tu* science, as evidenced by Project 523, forces us to confront how different valuations of what constituted science during Cultural Revolution generated both epistemological and methodological possibilities otherwise foreclosed.

Thinking Through Variances with *Changshan* Research

To understand how specific practices and approaches integral to Project 523 reflected different political valuations of science as *tu* science, it is worth setting up a point of comparison. In the 1940s, as Sean Lei has argued with respect to Nationalist medical research on *changshan* (常山), how one studied Chinese drugs articulated different political commitments about the relationship between Chinese medicine and modernity.⁷ *Changshan* is a febrifuge made from a hydrangea that has a long reference record going back to the earliest materia medica, the *Canon of the Divine Husbandman’s Materia Medica* (*Shennong bencaojing*, compiled in the first or second century AD). Although the efficacy of *changshan* could be extrapolated from a literature review of Chinese prescriptions for the treatment of malaria—*changshan* was an ingredient in one of the four major traditional methods for treating malaria—practitioners of Western medicine were hesitant to trust Chinese drugs “unless they had successfully been ‘translated’ into the socio-technical network of biomedicine.”⁸ Translation, however, required a process of re-networking in which Chinese drugs were disassociated from their traditional network

² An intriguing comparison can be made between Tu Youyou and Yuan Longping. Although the latter has become celebrated as China’s most famous agricultural scientist for his invention of hybrid rice, similar concerns have also been raised and would likely take a similar form should Yuan Longping ever be awarded the Nobel Prize. See Sigrid Schmalzer’s essay, “Yuan Longping, Hybrid Rice, and the Meaning of Science in the Cultural Revolution and Beyond,” in this issue. I want to thank Sigrid for bringing this point to my attention.

³ See, for example, “Zhongguo dalu kexuejia shou huo Nuobei’er kexuejiang xueshu taolun’ zhuanlan,” *Ziran bianzhengfa tongxun* (January 2016), <http://www.cnki.com.cn/Article/CJFDTotal-ZRBT201601001.htm>.

⁴ The issue of attribution is complicated. The social and political climate during the Cultural Revolution emphasized collective endeavor, and in the realm of scientific research, individual authorship was neither esteemed nor practiced in publications. This applies to Tu Youyou, who was not individually named in the original Chinese publications on artemisinin. For a more detailed consideration of how authorship, single or collective, factors into shifting valuations of *tu* science, see Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago Press, 2016), 78–80.

⁵ This is not to suggest there has been little or no recognition of her work or its potential significance for traditional Chinese medicine. For a more positive assessment as well as supporters of her achievement, see, for example, Li Bin’s Q & A with six Chinese medical practitioners in “Zhongyiyao shi yige weida baoku,” in Rao Yi, Zhang Daqing, and Li Runhong, *Youyou you hao: Tu Youyou yu qinghaosu* (Beijing: Zhongguo kexue jishu chubanshe, 2015), 126–30.

⁶ The best treatment of *tulyang* binary can be found in Sigrid Schmalzer, *Red Revolution, Green Revolution: Scientific Farming in Socialist China* (Chicago: University of Chicago, 2016). Although the terminology is different, Miriam Gross explores similar terrain in her notion of “grassroots science.” Gross, however, juxtaposes “grassroots science” against “normative science,” which presumably involves standard scientific criteria, such as reproducibility or control groups. This distinction makes sense given her focus on Mao-era anti-schistosomiasis campaigns, but by this measure, the concept of “grassroots science” is insufficient for understanding artemisinin research, because the researchers involved were committed to and believed themselves to be engaged in “normative science.” Miriam Gross, *Farewell to the God of Plague: Chairman Mao’s Campaign to Deworm China* (Berkeley: University of California Press, 2016).

⁷ The People’s Republic of China was established in 1949. From 1912 until 1949, China was a constitutional republic under the leadership of the Nationalist party (also referred to as “KMT” or “GMD”).

⁸ Sean Hsiang-lin Lei, *Neither Donkey nor Horse: Medicine in the Struggle over China’s Modernity* (Chicago: University of Chicago Press, 2014), 201.

观众了解，但中国对屠呦呦及其成就的赞扬却一直吝啬，充满了不满和不悦，甚至在她于2015年获得诺贝尔医学奖之后也是如此。国际社会对屠呦呦的赞扬引发了国内的关注和批评，认为她不应该独自获得如此多的赞誉。²2017年1月9日，她获得了中国最高科学荣誉——国家最高科学技术奖，这可以被视为官方对其成就的认可，但这并未平息各种不同意见的表达。有人问道，一个“三无”（san wu）科学家——即没有研究生学位、没有海外培训、也不是中国科学院院士的人——怎么能成为中国医学成就的代表人物？³有些人质疑她作为发现者的角色。⁴甚至她发现的故事也因被认为过分强调个人功绩而未能承认其他相关科学家的贡献而遭到强烈批评。⁵这种明显的困惑和自我反思表明，改革开放时代的中国科学——无论其表面上多么像全球科学——都保留着我们需要认真对待的独特特质。

本文探讨了青蒿素的发现如何阐明了科学价值观在不同时间和空间中的多样性和可塑性。将青蒿素研究置于更广阔的时间框架中，我们可以看到青蒿等中药如何通过其对科学话语中与“土”和“洋”相关的交替意义群的体现，为科学实验和创新开辟了空间。毛泽东时代的科学和政治话语经常强调“土”的重要性，而非“洋”。前者与“本土”、“中国”、“地方”、“乡土”、“大众”、“粗糙”等词语相关联，阐明了一种为社会主义革命理想服务的激进科学愿景。广大群众将在科学知识的生产中发挥形成性作用，通过“土”科学，毛泽东时代的中国力求实现其自力更生、

² An intriguing comparison can be made between Tu Youyou and Yuan Longping. Although the latter has become celebrated as China’s most famous agricultural scientist for his invention of hybrid rice, similar concerns have also been raised and would likely take a similar form should Yuan Longping ever be awarded the Nobel Prize. See Sigrid Schmalzer’s essay, “Yuan Longping, Hybrid Rice, and the Meaning of Science in the Cultural Revolution and Beyond,” in this issue. I want to thank Sigrid for bringing this point to my attention.

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群众动员和实际应用方面的承诺。⁶相比之下，“洋”则代表着“外国”、“西方”、“精英”、“专业”等特质，“洋”科学往往带有专业化、教育和培训方面的跨国网络以及对基础研究的重视等特点。

因为毛泽东时代药物研究所涉及的科学工作跨越了多个领域：实验室医学研究、传统本草书籍和文集的文本分析，以及以公共卫生名义进行的民众动员，重新审视青蒿素研究的历史，不仅将其视为现代中国医学史的一部分，更将其视为现代中国科学史不可或缺的一部分，这为我们提供了一个独特的视角，来思考中国科学、技术和医学（STM）与文化大革命之间的关系。事实上，对土科学（tu science）的更深入探索，正如523项目所证明的那样，迫使我们正视在文化大革命期间，对科学构成要素的不同评价是如何产生了原本被排除在外的认识论和方法论可能性。

通过常山研究思考差异

为了理解523项目中具体的实践和方法如何反映了将科学视为土科学的不同政治评价，有必要建立一个比较点。正如雷祥麟（Sean Lei）在谈到国民党对常山（）的医学研究时所指出的，在20世纪40年代，如何研究中药，表达了对中医药现代性之间关系的不同政治承诺。⁷常山是一种由绣球花制成的退热剂，其记载历史悠久，可追溯到最早的本草著作《神农本草经》（*Shennong bencaojing*, 公元一或二世纪编纂）。尽管常山的疗效可以从治疗疟疾的中文处方文献综述中推断出来——常山是治疗疟疾的四种主要传统方法之一的成分——但西方医学的实践者不愿相信中药，“除非它们成功地‘转化’为生物医学的社会技术网络。”⁸然而，这种转化需要一个重新组网的过程，其中中药与它们的传统网络脱离。

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⁸ Sean Hsiang-lin Lei, *Neither Donkey nor Horse: Medicine in the Struggle over China’s Modernity* (Chicago: University of Chicago Press, 2014), 201.

and assimilated into the socio-technical network of bio-medicine. In the case of *changshan*, a clinical experiment on human subjects, as conducted by Cheng Peizhen, verified the historically transmitted curative efficacy of *changshan*. But rather than relying on patients’ reports of their physical condition, Dr. Cheng measured the level of malaria-causing plasmodia in the patients’ blood.⁹ Only with this demonstration in place was it then possible for *changshan* to be picked up and investigated by different members of the socio-technical network of biomedicine.

Even though *changshan* had been successfully “translated,” the sequence of scientific steps taken aroused controversy. As Lei explains, the scientists involved in *changshan* research “recognized that their approach was generally perceived to be violating an ethical consensus in the scientific community,” especially with regard to experimentation on human subjects.¹⁰ Scientific convention for pharmaceutical research prescribed an order of research procedures that began with chemical analysis to identify the active component that was supposed to be responsible for the action of the drug. Once an isolated chemical had been identified, scientists could then proceed to conduct pharmacological experiments with laboratory animals—the success of which would determine the next step, clinical experiments with human subjects. With the clinical efficacy confirmed on humans, scientists could then attempt to synthesize the active component by artificial means and eventually develop a drug for production on an industrial scale. Research on *changshan* occurred in a manner that disrupted the prescribed sequence. Instead of beginning with a chemical analysis and proceeding through the listed steps, *changshan* research began with a clinical application (for instance, Cheng Peizhen’s clinical study of fifty students reportedly sick with malaria), and then based on those results moved “backwards,” as it were, through the various forms of analyses. By not adhering to the conventional order of research, biomedical scientists involved in *changshan* research had to justify their decision without ceding political and epistemological concessions to practitioners of Chinese medicine.

During the Republican period, practitioners of Chinese and Western medicine vied for alliance with the modernizing Chinese state. Because research protocols functioned as political strategies in this medical struggle, practitioners of Western medicine policed the boundaries of scientific value with vigilance. Any recognition that the conventional research protocol could or should be reordered when studying Chinese drugs risked unintentionally legitimating the scientific claims made by practitioners of Chinese medicine, who argued that use of clinical trials could validate the therapeutic efficacy of their “experience with the human body.”¹¹ Moreover, the adoption of a “clinical experiment first” strategy opened up possibilities

for ensuring that doctors of Chinese medicine could retain some degree of authority over the scientific reinscription and translation process. Clinical experimentation was in some measure the closest approximation to already existing forms of experimentation through individual use and past experience. In contrast, a research project that adhered to the conventional research program and began with chemical analysis would produce an extract that would be completely foreign to Chinese doctors. In effect, the resulting extract would constitute a fundamentally different object without obvious ties to the larger epistemic world and body of practices associated with Chinese medicines.¹² As we will see with Project 523, shifts in political tenor rearticulated the spectrum and ordering of what constituted scientific procedures. In light of the public affirmation of self-reliance, mass participation, and indigenous knowledge during the Cultural Revolution, Project 523’s malarial research adopted procedural role reversals and crossed disciplinary borders that were strictly discouraged, if not prohibited, for *changshan* research.

Project 523

How did the discovery of artemisinin come about? Like other Mao-era scientific endeavors, Project 523, which derived its name from the date (May 23, 1967) on which the National Commission of Science and Technology and the General Logistics Department of the Chinese People’s Liberation Army (PLA) convened a meeting to coordinate research on the prevention and treatment of malaria, included a heavy dose of military supervision, massive mobilization of human resources, ideological fervor, and secrecy. In 1964, North Vietnamese leaders requested medical assistance from the People’s Republic of China (PRC) in their battle with malaria. Fighting between American and North Vietnamese forces throughout the 1960s had been escalating, but it was malaria that especially imperiled soldier health for both sides. The increasing number of chloroquine-resistant malaria cases identified amongst the North Vietnamese civilian population further heightened governmental concern. When North Vietnamese leaders met with Mao Zedong, they spoke of the increasing danger posed by malaria and asked Mao for Chinese assistance in solving the malaria problem. Mao reportedly responded, “Solving your problem is

¹² Ibid., 5. Lei also makes the important point that the Received Program presupposed that the efficacy of Chinese drugs can be traced back to a single active chemical. This kind of reductionist framework is embedded in the very practices and procedures used to study Chinese drugs. Commitment to the identification of single active agents from plants may, however, be an unnecessarily narrow approach, especially when the botanical materials being studied are themselves drawn from bodies of knowledge and practices that highlight not an individual active agent, but the effects produced from its combination with other materials. As Lei and Gerard Bodecker argue, there may be “intended therapeutic synergies between the multiple constituents of a plant and among the many plants contained in traditional complex mixtures.” The biomedical approach to drugs has not historically recognized this possibility in its methodology. See Lei and Bodeker, “Changshan (*Dichroa febrifuga*)—Ancient Febrifuge and Modern Antimalarial: Lessons for Research from a Forgotten Tale,” in *Traditional Medicinal Plants and Malaria*, ed. Merlin Wilcox, Gerald Bodeker, and Philippe Rasoanaivo (Boca Raton, FL: CRC Press, 2004), 69–93.

⁹ Ibid., 205.

¹⁰ Ibid., 209. For a nuanced examination of the public and professional debates over human experimentation in the United States during the first half of the twentieth century, see Susan Lederer, *Subjected to Science: Human Experimentation in America before the Second World War* (Baltimore: John Hopkins University Press, 1995).

¹¹ Ibid., 93.

并被生物医学的社会技术网络所吸收。就常山而言，程培珍对人体进行的临床实验证实了常山的历史疗效。但程医生没有依赖患者对其身体状况的报告，而是测量了患者血液中引起疟疾的疟原虫水平。⁹只有在这一论证到位后，常山才有可能被生物医学社会技术网络的不同成员所采纳和研究。

尽管常山已成功“转化”，但所采取的科学步骤顺序却引起了争议。正如雷所解释的，参与常山研究的科学家们“认识到他们的方法普遍被认为违反了科学界的伦理共识”，尤其是在人体实验方面。¹⁰ 药物研究的科学惯例规定了一套研究程序，首先是化学分析，以确定被认为是药物作用的活性成分。一旦确定了分离出的化学物质，科学家就可以进行实验室动物的药理实验——其成功与否将决定下一步，即人体临床实验。在人体临床疗效得到证实后，科学家就可以尝试通过人工方法合成活性成分，并最终开发出可工业化生产的药物。对常山的研究以一种打破既定顺序的方式进行。常山研究没有从化学分析开始并按列出的步骤进行，而是从临床应用开始（例如，程培珍对据称患有疟疾的50名学生的临床研究），然后根据这些结果“倒退”进行各种形式的分析。由于不遵守传统的研究顺序，参与常山研究的生物医学科学家必须在不向中医药从业者做出政治和认识论让步的情况下，证明他们的决定是合理的。

民国时期，中西医从业者争相与现代化中的中国国家结盟。由于研究方案在医疗斗争中充当政治策略，西医从业者警惕地维护着科学价值的边界。任何承认在研究中药时可能或应该重新安排常规研究方案的做法，都可能无意中使中医师提出的科学主张合法化，他们认为临床试验可以验证其“人体经验”的治疗效果。¹¹ 此外，“临床实验优先”策略的采用开辟了可能性

⁹ 同上，205。¹⁰ 同上，209。关于20世纪上半叶美国人体实验的公众和专业辩论的细致考察，参见Susan Lederer, 《受制于科学：二战前美国的人体实验》（巴尔的摩：约翰·霍普金斯大学出版社，1995）。¹¹ 同上，93。

确保中医师在科学再铭写和转化过程中保留一定程度的权威。临床实验在某种程度上最接近于通过个体使用和过往经验进行的现有实验形式。相比之下，一个遵循常规研究计划并从化学分析开始的研究项目，将产生一种对中医医生来说完全陌生的提取物。实际上，由此产生的提取物将构成一个根本不同的客体，与中药相关的更广泛的认知世界和实践体系没有明显的联系。¹² 正如我们将在523项目中看到的那样，政治氛围的变化重新阐明了构成科学程序的范围和顺序。鉴于文化大革命期间对自力更生、群众参与和本土知识的公开肯定，523项目的疟疾研究采用了程序上的角色逆转，并跨越了对常山研究严格禁止（如果不是完全禁止）的学科界限。

523项目
青蒿素是如何被发现的？与毛泽东时代的其他科学研究一样，523项目（其名称来源于1967年5月23日，即国家科学技术委员会和中国人民解放军总后勤部召开会议协调疟疾防治研究的日期）包含了大量的军事监督、大规模的人力动员、意识形态热情和保密性。1964年，北越领导人请求中华人民共和国（PRC）提供医疗援助，以对抗疟疾。20世纪60年代，美国和北越军队之间的战斗不断升级，但疟疾尤其危及双方士兵的健康。在北越平民中发现的氯喹耐药疟疾病例数量不断增加，进一步加剧了政府的担忧。当北越领导人会见毛泽东时，他们谈到了疟疾日益增长的危险，并请求毛泽东帮助解决疟疾问题。据报道，毛泽东回应说：“解决你们的问题就是

¹² 同上，5。雷还提出了一个重要观点，即“既定方案”预设了中药的疗效可以追溯到单一活性化学物质。这种还原论框架根植于研究中药所使用的实践和程序中。然而，致力于从植物中识别单一活性物质可能是一种不必要的狭隘方法，特别是当所研究的植物材料本身来源于强调的不是单一活性物质，而是其与其他物质结合所产生的效果的知识体系和实践时。正如雷和杰拉德·博德克所说，在“植物的多种成分之间以及传统复杂混合物中包含的许多植物之间，可能存在预期的治疗协同作用”。生物医学的药物研究方法在历史上并未在其方法论中认识到这种可能性。参见Lei and Bodeker, “常山（*Dichroa febrifuga*）——古老的退热剂和现代抗疟药：从被遗忘的故事中汲取研究教训”，载于Merlin Wilcox, Gerald Bodeker和Philippe Rasoanaivo主编的《传统药用植物与疟疾》（佛罗里达州博卡拉顿：CRC出版社，2004），69-93页。

the same as solving our own.”¹³ The PLA initiated research late in 1964 on three existing drugs (pyrimethamine, dapsone, and sulfadoxine), and while modestly successful in developing two new combination antimalarial drugs, a decision was made that more resources—indeed, the resources of the entire nation—were needed to expedite research and achieve victory, hence Project 523.¹⁴

Although often treated as a colorful detail that underscores the stark political divisions distinctive of the Cold War, North Vietnam’s request and China’s acquiescence flag both the transnational dimensions of Chinese medical research during the Mao years and its incorporation of *tu* science.¹⁵ Scholars like Laurence Schneider, Gao Xi, Zhipeng Gao, and Zuoyue Wang have demonstrated the importance of transnational influence—most commonly in the form of Soviet experts and scientific orthodoxy after 1949 until 1960, but also through the employment of a largely Western-trained, elite scientific and technological Chinese work force—on Chinese science and technology.¹⁶ But what has been less well articulated about these transnational currents were how they refracted Chinese pride in the production of their own science. As Sigrid Schmalzer has written elsewhere, “In the context of the Cold War, Maoist ‘self-reliant science’ [*tu* science] was meant to bolster domestic confidence in Chinese socialist science and also to offer an alternative model for Third World countries.”¹⁷

As a potential scientific model, Project 523 knitted together recent experiences in mass participation/innovation and laboratory-based researches to achieve goals resonant with both domestic and allied concerns. On the

home front, malaria had already been designated in 1956 as one of the five parasitic diseases (schistosomiasis, malaria, filariasis, kala-azar, and ancylostomiasis) most detrimental to the people’s health.¹⁸ The fight against malaria encapsulated the four key principles guiding national health policy: serve the needs of workers, peasants, and soldiers; emphasize prevention as the first priority; unite Chinese and Western medicines; and combine health work with mass movements. A special national antimalarial conference, which ambitiously declared the intention of eliminating malaria within seven years, was convened in 1957.¹⁹ It marked the launch of a nation-wide antimalarial campaign that combined a range of measures, including the creation of county/township/village level malaria prevention and treatment stations in endemic areas to conduct investigations, train personnel, and set up pilot programs at designated sites to experiment with specific antimalarial strategies.²⁰

Coupled with this institutional network for prevention and surveillance were mass mobilization campaigns in antimalarial work. Mass campaigns throughout the 1950s and 1960s mobilized broad swathes of the rural population. Local communities worked together to destroy breeding places of mosquitoes, spray insecticides to kill adult mosquitoes, and eliminate common mosquito habitats by filling in holes and pits, draining ditches and water courses, diverting stagnant water, rebuilding or improving sewage systems, and removing vegetation along embankments and water surfaces.²¹ Public health campaigns sometimes incorporated the use of modern antimalarial drugs, but given both the scarcity and expense associated with such drugs, the Chinese state encouraged the use of native herbs and acupuncture in the prevention and treatment of malaria (Figure 1).²²

Even though the infrastructure and personnel organizing Project 523 relied heavily on biomedical theory and

¹⁸ Schistosomiasis is parasitic disease caused by a species of trematodes belonging to the genus *schistosoma* (the snail is the transmission vector); filariasis is a parasitic disease caused by roundworms; kala-azar also known as “black fever” or DumDum fever is a severe form of leishmaniasis, a disease caused by the protozoan parasites of the *leishmania* genus; and ancylostomiasis is a parasitic disease caused by Ancylostoma hookworms.

¹⁹ It is conceivable that the Chinese intention to eliminate malaria in seven years refracted the World Health Organization’s global ambition to eradicate malaria, a program that was launched in 1955 to the cost of nearly \$1.4 billion over the subsequent fourteen years. Further research would need to examine if and how these programs were in dialogue and the extent to which the Chinese campaigns against malaria were communicated to non-Chinese audiences as effective, third way solutions to public health crises. At the very least, it is worth pointing out one crucial difference between these two endeavors. Ka-che Yip has highlighted that the Chinese approach did not rely on the near exclusive use of DDT, and instead attempted to build an “effective examination and surveillance system . . . [and] an anti-epidemic network” to achieve malaria elimination (*xiaomei*). Ka-che Yip, “Antimalarial Work in China: A Historical Perspective,” *Parassitologia* 40 (1998): 29–38, on 33, and Randall Packard, *The Making of a Tropical Disease: A Short History of Malaria* (Baltimore: John Hopkins University Press, 2007). I thank Hilary Smith for raising the WHO question.

²⁰ Yip, “Antimalarial Work in China” (ref. 19).

²¹ Liping Bu, “Anti-Malaria Campaigns and the Socialist Reconstruction of China, 1950–80,” *East Asian History* 39 (2014): 117–30; Yip, “Antimalarial Work in China” (ref. 19), 33–37; and Ka-che Yip, “Disease, Society, and the State: Malaria and Health Care in Mainland China,” in *Disease, Colonialism, and the State: Malaria in Modern East Asian History*, ed. Ka-che Yip (Hong Kong: Hong Kong University Press, 2009), 103–20.

²² Certain pharmaceuticals like penicillin and streptomycin did become both more affordable and prevalent over the course of the 1960s and 1970s. See Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester: University of Rochester Press, 2012), chs. 3 and 4.

与解决我们自己的问题一样。”¹³解放军于1964年末开始研究三种现有药物（乙胺嘧啶、氨苯砜和磺胺多辛），虽然在开发两种新型复方抗疟药方面取得了适度成功，但仍决定需要更多资源—事实上，是全国的资源—来加速研究并取得胜利，因此启动了523项目。¹⁴

尽管这常被视为一个生动的细节，突显了冷战时期鲜明的政治分歧，但北越的请求和中国的默许都标志着毛泽东时代中国医学研究的跨国维度及其对本土科学的融合。¹⁵劳伦斯·施耐德、高晞、高志鹏和王作跃等学者已经证明了跨国影响的重要性—最常见的是1949年至1960年间以苏联专家和科学正统形式出现，但也通过雇佣一支主要受西方培训的精英科学技术中国劳动力—对中国科学技术的影响。¹⁶

然而，这些跨国潮流中，中国对其自身科学成果的自豪感是如何折射出来的，这一点却鲜有阐述。正如西格丽德·施马尔策在其他地方所写：“在冷战背景下，毛泽东时代的‘自力更生科学’ [土科学] 旨在增强中国社会主义科学的国内信心，并为第三世界国家提供一种替代模式。”¹⁷

作为一种潜在的科学模式，523项目将大规模参与/创新和实验室研究的近期经验结合起来，以实现与国内和盟友关切相符的目标。在国内方面，

¹³ Quoted in Li Runhong, “‘523 renwu’ yu qinghao kangnue zuoyong de zai faxian,” *Zhongguo keji shi zazhi* [*Chinese Journal of the History of Science and Technology*] 32, no. 4 (2011): 489. Malaria was not a problem for the Vietnamese alone. There appears to be some variance in the statistics, but the overall conclusion remains the same. Christine Beadle and Stephen L. Hoffman report in their “History of Malaria in the United States Naval Forces at War: World War I through the Vietnam Conflict,” that within an eight-year period between 1964 and 1973, there were 24,606 cases of malaria, an estimated 391,965 sick days because of malaria, and forty-six deaths due to malaria. By contrast, during the Korean War there were 4542 diagnoses of malaria, 50,294 sick-days due to malaria, and no deaths in the naval forces. Lynn W. Kitchen, David W. Vaughn, and Donald R. Skillman cite some 81,000 cases of malaria, with 1.4 million malarial sick days, and 133 deaths. The difference may stem from Beadle and Hoffman’s focus on the naval forces alone. Military concern over the debilitating effects of malaria were longstanding, and during the Vietnam War, the situation was aggravated by the emergence of chloroquine-resistant malaria and insecticide-resistant mosquitoes. See Beadle and Hoffman, “History of Malaria in the United States Naval Forces at War,” *Clinical Infectious Diseases* 16, no. 2 (1993): 320–29, on 326; and Kitchen, Vaughn, and Skillman, “Role of US Military Research Programs in the Development of US Food and Drug Administration-Approved Antimalarial Drugs,” *Clinical Infectious Diseases* 43, no. 1 (2006): 67–71, on 69.

¹⁴ Zhang Jianfang, *Wuersan xiangmu yu Qinghaosu yanfa jishi* [*A Detailed Chronological Record of Project 523 and the Discovery and Development of Qinghaosu [Artemisinin]*] (Guangzhou: Yangcheng wanbao chubanshe, 2006), 14–15.

¹⁵ If viewed from the perspective of the PRC, the stark political divisions of the Cold War in 1964 entailed not only antagonism with the United States, but also with Soviet Union. The break in Sino-Soviet relations occurred in 1960.

¹⁶ 关于20世纪50年代跨国对中国科学的影响的讨论，参见高晞，《向苏联学习：巴甫洛夫对中国医学的影响，20世纪50年代》，载于卜立平、叶嘉智主编《战后亚洲的公共卫生与国家重建：国际影响与地方转型》（牛津：劳特利奇，2015），第72-89页；高志鹏，《中国的巴甫洛夫主义：毛泽东时代科学学科的政治与分化》，载于《科学史》第53卷第1期（2015）：第57-85页；劳伦斯·A·施耐德，《20世纪中国的生物学与革命》（兰哈姆：罗曼与利特菲尔德，2003），第115-212页；王作跃，《冷战时期中国的发展型国家：1956年十二年科学技术规划的制定》，载于《历史与技术》第31卷第3期（2015）：第180-205页。¹⁷

Sigrid Schmalzer, “自力更生科学：冷战对社会主义中国科学的影响”，载于 Naomi Oreskes 和 John Kriege 主编，《全球冷战中的科学与技术》（马萨诸塞州剑桥：麻省理工学院出版社，2014），第 79 页。

在国内方面，疟疾在1956年已被指定为对人民健康危害最大的五种寄生虫病（血吸虫病、疟疾、丝虫病、黑热病和钩虫病）之一。¹⁸ 抗击疟疾的斗争体现了指导国家卫生政策的四个关键原则：为工人、农民和士兵的需求服务；强调预防是首要任务；中西医结合；以及将卫生工作群众运动相结合。1957年召开了一次全国性的抗疟疾特别会议，雄心勃勃地宣布了在七年内消灭疟疾的意图。¹⁹ 这标志着一场全国性抗疟疾运动的启动，该运动结合了一系列措施，包括在流行地区建立县/乡/村级疟疾防治站，进行调查、培训人员，并在指定地点设立试点项目，以试验特定的抗疟疾策略。²⁰

与这种用于预防和监测的机构网络相结合的是抗疟疾工作中的群众动员运动。整个20世纪50年代和60年代的群众运动动员了广大农村人口。当地社区共同努力，通过填补坑洼、排干沟渠和水道、疏导死水、重建或改善污水处理系统以及清除堤岸和水面上的植被来消灭蚊子滋生地、喷洒杀虫剂杀死成年蚊子以及消除常见的蚊子栖息地。²¹ 公共卫生运动有时会结合使用现代抗疟疾药物，但考虑到此类药物的稀缺性和昂贵性，中国政府鼓励使用本土草药和针灸来预防和治疗疟疾（图1）。²²

尽管组织523项目的基建和人员严重依赖生物医学理论和

¹⁸ Schistosomiasis is parasitic disease caused by a species of trematodes belonging to the genus *schistosoma* (the snail is the transmission vector); filariasis is a parasitic disease caused by roundworms; kala-azar also known as “black fever” or DumDum fever is a severe form of leishmaniasis, a disease caused by the protozoan parasites of the *leishmania* genus; and ancylostomiasis is a parasitic disease caused by Ancylostoma hookworms.

¹⁹ It is conceivable that the Chinese intention to eliminate malaria in seven years refracted the World Health Organization’s global ambition to eradicate malaria, a program that was launched in 1955 to the cost of nearly \$1.4 billion over the subsequent fourteen years. Further research would need to examine if and how these programs were in dialogue and the extent to which the Chinese campaigns against malaria were communicated to non-Chinese audiences as effective, third way solutions to public health crises. At the very least, it is worth pointing out one crucial difference between these two endeavors. Ka-che Yip has highlighted that the Chinese approach did not rely on the near exclusive use of DDT, and instead attempted to build an “effective examination and surveillance system . . . [and] an anti-epidemic network” to achieve malaria elimination (*xiaomei*). Ka-che Yip, “Antimalarial Work in China: A Historical Perspective,” *Parassitologia* 40 (1998): 29–38, on 33, and Randall Packard, *The Making of a Tropical Disease: A Short History of Malaria* (Baltimore: John Hopkins University Press, 2007). I thank Hilary Smith for raising the WHO question.

²⁰ Yi, “Antimalarial Work in China” (ref. 19); 重建, 1950–80”, 《东亚历史》39 (2014): 117–30; Yip, “中国的抗疟工作”（参考文献 19），33–37；以及 Ka-che Yip, “疾病、社会与国家：中国大陆的疟疾与医疗”，载于 Ka-che Yip 主编，《疾病、殖民主义与国家：现代东亚历史上的疟疾》（香港：香港大学出版社，2009），103–20。

²² Certain pharmaceuticals like penicillin and streptomycin did become both more affordable and prevalent over the course of the 1960s and 1970s. See Xiaoping Fang, *Barefoot Doctors and Western Medicine in China* (Rochester: University of Rochester Press, 2012), chs. 3 and 4.



Figure 1. “We are determined to eradicate malaria,” produced by the Revolutionary Committee of the Health and Epidemic Prevention Station of Jinan City, May 1970. Source: Chinese Public Health Collection, National Library of Medicine, National Institutes of Health, Bethesda MD, USA.

praxis, Chinese medicine occupied an important place in its various research activities. Ideological celebration of, as well as practical necessity for, self-reliance and a genuine ethos of mutual assistance facilitated the general opening of the boundaries that had hitherto defined Chinese medicine. Folk remedies (*tufang* 土方) and secret recipes (*mifang* 秘方) were actively sought out; new acupuncture points and treatment techniques were devised.²³

²³ See, for example, “Ba jilei de jingyan he zuchuan mifang chuanshou gei qingnian yidai: Guangzhoushi guke lao Zhongyi He Zhulin rexin dai tudi,” *Renmin ribao* (December 15, 1961) and Pan Di, “Qisi huisheng tuyang qifang yicai, shejiweiren xin de pinzhi chengzhang,” *Renmin ribao* (September 9, 1958).



图1. 1970年5月济南市卫生防疫站革命委员会制作的“我们决心消灭疟疾”。来源：美国国立卫生研究院国家医学图书馆中国公共卫生藏品，马里兰州贝塞斯达。

实践中，中医药在其各项研究活动中占据了重要地位。对自力更生的思想上的推崇以及实际上的必要性，以及真正的互助精神，促进了此前界定中医药的界限的普遍开放。积极寻求民间疗法（土方）和秘方（秘方）；设计了新的针灸穴位和治疗技术。²³ 土方

秘方

²³ 例如，参见《人民日报》（1961年12月15日）刊登的“把积类的经验和祖传秘方传授给青年一代：广州市骨科老中医何竹林热心带徒弟”以及潘涤在《人民日报》（1958年9月9日）刊登的“起死回生土洋奇方异彩，设计为人新的品质成长”。

解放军523项目研究所制定的最初三年计划旨在“远近结合，中西医结合，以药为主，重在创新，统一计划，分工合作”。²⁴军民科研协作的目的，涉及全国60多个工作单位、多个部门和关键科技人员，正是为此而设计的。

²⁴ Zhang Jianfang, *Wuersan xiangmu yu Qinghaosu yanfa jishi* (ref. 14), 15.

Within weeks of formation, Project 523 subgroups had initiated systematic observations of the efficacy of compound prescriptions for antimalarial prevention, textual investigations of Chinese materia medica, public surveys to identify popular folk remedies, collections of local secret or tried-and-true recipes, and organized expeditions to gather the various kinds of plant, animal, and mineral material for Chinese drugs. All of these activities presumed a level of mass participation that only made sense within the highly politically charged context of Maoist China.

Project 523's decision to investigate Chinese drugs through laboratory research was not in and of itself unprecedented. When Chen Kehui and Carl Schmidt of the Peking Union Medical College coauthored and published their original paper on *mahuang* in 1924, they ignited a research fire such that in the five years subsequent, more than 500 research papers on ephedrine (for relief for asthma) appeared around the world. In the 1940s, state interest in the Chinese drug *changshan* for its antimalarial properties led to the establishment of the Nationalist government-funded Research Institute for Chinese Specific Drugs and the Jingfo Mountain Chinese Drug Experimental Farm in Sichuan.

If scientific interest in Chinese drugs was not new, the institutional climate within which Project 523 investigators worked was nonetheless different in important ways from earlier antimalarial research efforts. The Vietnam War had exacerbated an epidemiological crisis to which Maoist China responded with nationalist fervor by turning to its institutions of traditional Chinese medicine. That such institutions were in the 1960s a newly invented, mixing ground of specialists—many of whom possessed more than a passing familiarity with both Chinese medicine and biomedicine—stemmed from the radical policy adopted by the Chinese Communist Party between 1954 and 1956 to integrate Chinese medicine and Western medicine (*zhongxiyi jiehe* 中西醫結合).²⁵ As Fu Lianzhang, the president of the Chinese Medical Association and vice-minister of the Ministry of Health explained in a 1954 article, the objective was to create a new medicine—“This One Medicine will possess a basis in the natural sciences, will have absorbed the ancient and the new, the Chinese and the foreign, all medical achievements—and will be China's New Medicine.”²⁶

Among its first initiatives, the Maoist state instituted a program in which doctors of Western medicine or biomedicine had to study Chinese medicine (*xiyi xuexi zhongyi* 西醫學習中醫). In Mao's words, “The study of Chinese medicine by Western medical practitioners is honorable. This is because through the studying and improving [of Chinese medicine], the demarcation lines between Chinese and Western medicine can be discarded, to form one Unified Medicine of China, as a contribution to the whole world.”²⁷ In what many perceived as a role reversal,



Figure 2. Tu Youyou (right) with her teacher Lou Zhicen (left) in 1951. Source: Wikipedia.

Chinese medicine was elevated to a position of comparative superiority. At a moment in which the confidence of the CCP, as well as its control over China, was growing, political affirmation for Chinese medicine coincided with a broader gesture towards reclaiming Chinese cultural heritage as not only a source of national pride, but also potential model of worldwide significance (Figure 2).²⁸

The ordering, that is, Western medicine learning from Chinese medicine, reflected both epistemological and political imperatives in the formation of this New Medicine as *tu* science. By forcing doctors of Western medicine to learn Chinese medicine, Mao especially hoped to control and rectify perceived undesirable ideological tendencies and institutional prerogatives of Western-trained doctors, while also diminishing dependency on imported technology and drugs. Values like self-reliance through the development of indigenous resources—material, methodological, and human—were explicitly invoked and imbued with greater political force.

This reevaluation of Chinese medicine led to enhanced institutional support and development.²⁹ Parallel yet intertwining institutional structures were created, and the diversity—and factionalism between contending traditions—of traditional medical thought was systematized and standardized through textbooks and training.³⁰ More than half of the graduating first class of “Western doctors study Chinese medicine” were assigned positions within the China Academy of Traditional Chinese Medicine,

²⁸ Other cultural forms being rehabilitated included architecture, painting, and theater. Mao, for one, was convinced that China's experience of, and ultimate success in achieving, a socialist revolution would serve as both inspiration and example for the world. “There are many countries that, like us, have suffered oppression of imperialism and feudalism, and [the people of these countries] constitute the majority of the world's population. It would be helpful to the people in these countries for us to have a revolutionary constitution, a people's democratic constitution, and a clear, precise, and correct road to follow.” Quoted in *ibid.*, 66.

²⁹ As Scheid and Karchmer have emphasized, bias against Chinese medicine within the state apparatus was curbed, but not eliminated, and in certain substantive ways that lay outside the purview of this essay, the place of Chinese medicine remained subservient to Western medicine. That said, the broader endeavor to create a New Medicine through the integration of Chinese and Western medicines can be fruitfully examined as part of a more systematic attempt to create *tu* science. Scheid and Karchmer, “History of Medicine, 1890–2010” (ref. 25), 164.

³⁰ Volker Scheid, *Chinese Medicine in Contemporary China: Plurality and Synthesis* (Durham, NC: Duke University Press, 2002), ch. 3.

在成立的几周内，523项目小组就已开始系统地观察复方抗疟处方的疗效，对中药材进行文献研究，进行公众调查以确定流行的民间疗法，收集当地秘方或久经考验的验方，并组织考察队收集各种植物、动物和矿物材料用于中药。所有这些活动都预设了只有在毛泽东时代高度政治化的背景下才有意义的大众参与程度。

523项目通过实验室研究调查中药的决定本身并非史无前例。1924年，北京协和医学院的陈克恢和卡尔·施密特合著并发表了他们关于麻黄的原始论文，引发了一场研究热潮，以至于在随后的五年中，全球出现了500多篇关于麻黄碱（用于缓解哮喘）的研究论文。20世纪40年代，国家对中药常山抗疟疾特性的兴趣促成了国民政府资助的中国特种药物研究所和四川金佛山中药实验农场的建立。

如果对中药的科学兴趣并非新鲜事，那么523项目研究人员所处的制度环境与早期的抗疟研究工作仍有重要不同。越南战争加剧了流行病危机，毛泽东时代的中国以民族主义热情回应，转向其传统中医机构。这些机构在1960年代是新近创建的、专家们（—其中许多人对中医药和生物医学都有相当的了解—）混合的场所，这源于中国共产党在1954年至1956年间采取的将中医药与西医药结合的激进政策（中西医结合）。²⁵正如中华医学会会长、卫生部副部长傅连璋在1954年的一篇文章中所解释的，目标是创造一种新医学—“这种新医学将以自然科学为基础，吸收古今中外所有医学成就—并将成为中国的新医学。”²⁶ 中西醫結合

作为其首批举措之一，毛泽东时代的国家设立了一项计划，要求西医或生物医学医生学习中医（*xiyi xuexi zhongyi* 西醫學習中醫）。用毛泽东的话说，“西医学习中医是光荣的。这是因为通过学习和改进[中医]，，可以消除中西医之间的界限，形成一个统一的中国医学，为全世界做出贡献。”²⁷ 在许多人看来，这是一种角色逆转，

²⁵ Volker Scheid and Eric I. Karchmer, “医学史, 1890-2010”, 载于《现代中国宗教II, 1850-2015》, 第1卷, Vincent Goossaert, Jan Kiely和John Lagerwey编（莱顿: Brill, 2016）, 141-96页。²⁶ Kim Taylor, 《早期共产主义中国的中国医学, 1945-63: 一场革命的医学》（伦敦: RoutledgeCurzon, 2005）, 第3章。引文来自73页。²⁷ 引自同上, 68页。



图2. 屠呦呦（右）与她的老师楼之岑（左）摄于1951年。来源：维基百科。

中医被提升到相对优越的地位。在中国共产党信心和对中国控制力不断增强的时刻，对中医的政治认同与更广泛的重新主张中国文化遗产的姿态相吻合，这不仅是民族自豪感的源泉，也是具有世界意义的潜在典范（图2）。²⁸

这种西医向中医学习的顺序，反映了这种作为“土科学”的新医学形成过程中的认识论和政治需要。通过强制西医学习中医，毛泽东尤其希望控制和纠正西方培训医生中被认为不理想的意识形态倾向和制度特权，同时减少对进口技术和药物的依赖。通过开发本土资源—物质、方法和人力—实现自力更生等价值观被明确提出，并被赋予更大的政治力量。

对中医的重新评估促使机构支持和发展得到加强。²⁹ 并行而又相互交织的制度结构得以建立，传统医学思想的多样性—以及相互竞争的传统之间的派系斗争—通过教科书和培训得到系统化和标准化。³⁰超过一半的“西医学习中医”首届毕业生被分配到中国中医科学院工作，

²⁸ Other cultural forms being rehabilitated included architecture, painting, and theater. Mao, for one, was convinced that China's experience of, and ultimate success in achieving, a socialist revolution would serve as both inspiration and example for the world. “There are many countries that, like us, have suffered oppression of imperialism and feudalism, and [the people of these countries] constitute the majority of the world's population. It would be helpful to the people in these countries for us to have a revolutionary constitution, a people's democratic constitution, and a clear, precise, and correct road to follow.” Quoted in *ibid.*, 66.

²⁹ As Scheid and Karchmer have emphasized, bias against Chinese medicine within the state apparatus was curbed, but not eliminated, and in certain substantive ways that lay outside the purview of this essay, the place of Chinese medicine remained subservient to Western medicine. That said, the broader endeavor to create a New Medicine through the integration of Chinese and Western medicines can be fruitfully examined as part of a more systematic attempt to create *tu* science. Scheid and Karchmer, “History of Medicine, 1890–2010” (ref. 25), 164.

³⁰ Volker Scheid, *Chinese Medicine in Contemporary China: Plurality and Synthesis* (Durham, NC: Duke University Press, 2002), ch. 3.

where traditional Chinese practitioners commingled with “hard-core scientists” like chemists, pharmacognosists, and pharmacologists.³¹ Within this mixed environment, *qinghao* research advanced among scientists, “who themselves had learnt the ways of appreciating traditional knowledge, [and who] worked side by side with historians of traditional medicine, who had textual learning.”³²

The values of *tu* science enabled researchers to pursue a program of research that recognized the empirical value of “unconventional” clinical experimentation (cf. *changshan* research). By deliberately seeking out native recipes and methods (*tufang tufa* 土方土法), researchers sanctioned the empirical value of what might otherwise be dismissed as anecdotal evidence and thereby legitimated a form of human experimentation as the entry point for drug research. In 1969, for example, the Project 523 Chinese medicinal herb research group in Nanjing documented extensive use by local residents of locally grown *qinghao* in both the prevention and treatment en masse of malaria. There was a common expression, “Don’t worry if you get malaria (*nüe ji* 瘧疾), just take *qinghao* with some brown sugar (*hongtang* 紅糖).”³³ The Jiaoshan production brigade (Jiaoshan dadui 焦山大隊) of the Longben Commune (Longben gongshe 龍奔公社) in Gaoyou county, Jiangsu province suffered from a high incidence rate of malaria. In 1969, the brigade initiated the use of *qinghao* in malaria prevention and treatment among its members and quickly transformed their malaria situation. They also implemented a multi-year observational study (1969–1972) of 184 malaria patients taking *qinghao*.³⁴ These on-site clinical experimentations with *qinghao* operationalized key hallmarks of *tu* science: self-reliance, mass mobilization, and nativism. Local communities mobilized local resources and attempted to adapt characteristically Chinese recipes and methods to achieve modern scientific and public health goals. Their efforts complemented the other coeval research activities undertaken by Project 523, be it obtaining ethanol extracts or conducting experiments on animal models.

In a similar fashion, the sort of boundary-policing that had characterized *changshan* research was both an anathema and politically dangerous during the Cultural Revolution. Doubting the therapeutic value of *qinghao*, or other *tufang*, would have been akin to doubting the wisdom of the masses. At the same time, because *tu* science endeavored to be a modern science, the epistemological value and logistical practicality associated with first evaluating clinically a Chinese drug’s curative efficacy was easily assimilated into the research program of Project 523. It became one step among many (textual, chemical, and clinical) that mapped out potential material for the development of new anti-malarial drugs. Moreover, Project 523 researchers did not limit these clinical experimentations to one phase of the overall project.

In 1972, a “Validating Group of *Qinghao* in Malaria Prevention and Treatment” (*Qinghao fangzhi nüejì*

yanzhengzu 青蒿防治瘧疾驗證組) was formed by members of the local Gaoyou county health department, health and quarantine station, and the Cheluo District Hospital. With a membership of 101 persons of a various expertise (practitioners of Western medicine, traditional Chinese medical doctors, and village doctors), the group set up eight clinical observation groups that tested the antimalarial effects of *qinghao* in different forms: fresh juice; a quick-boiled form; decoction; pills and tables; fresh versus dried *qinghao*; young thin *qinghao* branches and leaves versus thick branches and leaves; long versus short duration of boiling; and large versus small dose.³⁵ Their trials ran for several years, even after Tu Youyou and her team at the Beijing Institute of Materia Medica had obtained in October 1971 an initial crude extract of artemisinin (*qinghaosu* 青蒿素) that had a 100 percent success rate against *Plasmodium berghei* malaria in mice. When Tu Youyou reported their findings in March 1972 to the national Project 523 head office in Nanjing, it was decided that research on extraction and identification of the active component in *qinghao* should be expedited while clinical studies were simultaneously carried out. The ongoing nature of clinical experimentation with *qinghao* likely reflected exigencies of the period, in which readily available local solutions trumped reliance upon outside (including central government) resources. And yet, the willingness to pursue multiple research protocols at the same time suggests that *tu* science afforded a more expansive conception of scientific research. It authorized and legitimated clinical experimentation in ways consonant with Maoist ideals of self-reliance and the integration of research and extension.

Tu Medical Science

As Sigrid Schmalzer has pointed out, *tu* and *yang* mapped easily onto other, more famous binaries that structured Maoist approaches to science: red versus expert (that is, political commitment versus technical expertise); and theory versus practice. Characteristic of Maoist dialectical materialism and Cold War geopolitics, these binaries suggest a natural extension with the *zhong/xi* medical binary, with *tu* largely synonymous with *zhong* (Chinese) and *yang* with *xi* (Western). In many instances, *zhong/xi* and *tu/yang* appear as straightforward equivalents, whose respective discursive qualities nonetheless operated in different disciplinary domains. A general search of the *tu/yang* dyad in the *Renmin Ribao* (*The Peoples’ Daily*), the central government’s main media mouthpiece, suggests that *tu/yang* was more common to agricultural and industrial sciences. It was less commonly used in reference to medicine, where the *zhong/xi* dyad was more prominent. And yet, there was a clear parallelism between the two. In a manner not dissimilar to early PRC attempts to promote the “integration of Chinese and Western medicine” (*zhongxiyi jiehe*), the “integration of native/Chinese and foreign [science/technology]” (*tuyang jiehe*) emerged during the heady days of the Great Leap Forward (1958–62) as Mao launched the country on an unprecedented

³⁵ Zhang, *Wuersan xiangmu yu Qinghaosu yanfa jishi* (ref. 14), 21.

传统中医从业者与“核心科学家”如化学家、生药学家和药理学家融合在一起。³¹在这种混合环境中，青蒿研究在科学家中取得了进展，“他们自己学会了欣赏传统知识的方法，[并且]与传统医学史学家并肩工作，这些史学家拥有文本知识。”

³²

土科学的价值使研究人员能够开展一项研究计划，该计划认可“非传统”临床实验的经验价值（参见常山研究）。通过刻意寻找本土配方和方法（土方土法），研究人员认可了原本可能被视为轶事证据的经验价值，从而使一种人体实验形式合法化，作为药物研究的切入点。例如，1969年，南京523项目中国中草药研究小组记录了当地居民广泛使用当地种植的青蒿进行疟疾的预防和集体治疗。当时有一句俗语：“得了疟疾（疟疾）别担心，吃点青蒿加点红糖（红糖）就行。”³³江苏省高邮县龙奔公社（龙奔公社）焦山大队（焦山大队）疟疾发病率很高。1969年，该大队开始在其成员中推广使用青蒿进行疟疾预防和治疗，并迅速改变了他们的疟疾状况。他们还^{瘧疾紅糖}对184名服用青蒿的疟疾患者进行了为期多年的观察性研究（1969-1972年）。³⁴这些青蒿的现场临床实验使土科学的可行性得以实现：自力更生、群众动员和本土主义。当地社区动员当地资源，并试图将具有中国特色的配方和方法应用于实现现代科学和公共卫生目标。他们的努力补充了523项目同时进行的其他研究活动，无论是获取乙醇提取物还是进行动物模型实验。

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与此类似，在文化大革命期间，对常山研究的界限界定既是一种诅咒，也是一种政治危险。怀疑青蒿或其他土方的治疗价值，就如同怀疑群众的智慧。与此同时，由于土科学致力于成为一门现代科学，因此对中药的疗效进行首次临床评估所带来的认识论价值和后勤实用性，很容易被吸收到523项目的研究计划中。它成为绘制新抗疟药开发潜在材料的众多步骤（文本、化学和临床）之一。此外，523项目研究人员并未将这些临床实验局限于整个项目的一个阶段。

1972年，“青蒿防治疟疾验证组”成立

³¹ Elisabeth Hsu, “青蒿抗疟疾‘发现’的反思”，《英国临床药理学杂志》，61, 第6期（2006）：666–70, 第667页。³² 同上。³³ 张，五二三项目与青蒿素研发纪实（参考文献14），21。³⁴ 同上。

yanzhengzu 青蒿防治瘧疾驗證組) 由当地高邮县卫生部门、卫生检疫站和车逻区医院的成员组成。该小组由101名不同专业（西医、中医和乡村医生）的成员组成，设立了八个临床观察组，测试了青蒿不同形式的抗疟效果：鲜榨汁；速煮形式；煎剂；药丸和片剂；新鲜与干燥的青蒿；幼嫩的青蒿枝叶与粗壮的枝叶；长时间与短时间煮沸；大剂量与小剂量。³⁵

³²

³³

他们的试验持续了数年，甚至在屠呦呦和她在北京药物研究所的团队于1971年10月获得青蒿素（qinghaosu）的初步粗提物之后，该提取物在小鼠疟原虫疟疾中取得了100%的成功率。当屠呦呦于1972年3月向南京的全国523项目总部报告他们的发现时，决定加快青蒿中活性成分的提取和鉴定研究，同时进行临床研究。青蒿临床实验的持续进行可能反映了那个时期的紧迫性，即现成的本地解决方案胜过依赖外部（包括中央政府）资源。然而，同时进行多项研究方案的意愿表明，土科学提供了一种更广阔的科学研究概念。它以符合毛泽东自力更生和研究与推广相结合的理想的方式，授权并使临床实验合法化。

³²

³³

土医学

正如 Sigrid Schmalzer 所指出的，土和洋很容易映射到其他更著名的二元对立上，这些二元对立构成了毛泽东时代科学方法的基础：红与专（即政治承诺与技术专长）；以及理论与实践。作为毛泽东辩证唯物主义和冷战地缘政治的特征，这些二元对立与中/西医学二元对立自然延伸，其中土在很大程度上是中（中国）的同义词，洋是西（西方）的同义词。在许多情况下，中/西和土/洋似乎是直接的等价物，尽管它们各自的话语特质在不同的学科领域中发挥作用。《人民日报》（中央政府的主要媒体喉舌）对土/洋二元对立的一般搜索表明，土/洋在农业和工业科学中更为常见。它在医学中较少使用，医学中中/西二元对立更为突出。然而，两者之间存在明显的并行性。与中华人民共和国早期尝试推广“中西医结合”的方式不谋而合，“土洋结合”在大跃进（1958-1962）的狂热时期出现，当时毛泽东带领国家进行了一场前所未有的[科学/技术]运动。

³⁵ 张，五二三项目与青蒿素研发纪实（参考文献14），21。

social and economic campaign for rapid industrialization and collectivization.³⁶ Maoist aspirations for a science that neither disparaged nor negated mass participation and Chinese/native know-how also framed the ways in which medical research was approached and undertaken during the highly politicized period of the Cultural Revolution.

There were distinct ways in which *tu*, when aligned with medicine (*yi*), yielded subtly different connotations. *Tu* in the field of medical research could vary in inflection, ranging from folk and superstitious to local (as opposed to national) and indigenous. These articulations were enfolded in the broader social and political privileging of *tu* as evocative of self-reliance, mass participation, and practical experience. For this reason, *tu* could also gesture toward forms of empirical experience not usually recognized by medical science. In a curious way, the discovery of artemisinin illustrates how *tu* science legitimated a process of reassigning and re-networking the epistemological value of Chinese drugs that had previously been highly contentious in the 1940s. The notion of *tu* science allowed for the reordering of research protocols in a manner that did not compromise the scientificity of the results and prevent international recognition of the value of artemisinin as one of the most efficacious anti-malarial drugs available.

The case of artemisinin affords us an opportunity to think through how modern science in Cultural Revolution-era China might have worked under the banner of *tu* science. This is not to suggest that Mao era medical research possessed few, if any, *yang* characteristics; nor should one conclude that an emphasis on *tu* science necessarily entailed a rejection of *yang* science, although the two were in tension. Space does not permit a fuller examination of the *yang* aspects of medical research, but suffice it to say that the high degree of technical expertise expected of and exhibited by researchers, the on-going creation of specialized institutes for advanced training and teaching, the institutional organization of civilian and military research organs, and the general research protocols and equipment used all indicate the presence and importance of *yang* science. My interest in this essay has been to map out the effects of *tu* on *yang*. Put differently, an exploration of *tu* science, as evidenced by Project 523, reveals that different valuations of what constituted science in Maoist China generated both epistemological and methodological possibilities otherwise foreclosed, especially when we compare Project 523's scientific experimentation with Nationalist research on *changshan*.

Because *tu* and *yang* were deemed constituent elements of scientific research, I have argued that the space of possibility and scientific legitimacy operated differently during the Cultural Revolution period. Under the dictates of *tu* science, Chinese drugs, that is, traditional materia

medica, folk remedies, and secret recipes, and the experiential knowledge associated with their therapeutic efficacy, were elevated and esteemed. Although Maoist interest in Chinese drugs was not in itself new, the contentious struggle to separate and demarcate the epistemological value of Chinese drugs from other elements of traditional medicine that had characterized those earlier episodes was largely neutralized under the banner of *tu* science.³⁷ We see this most clearly by juxtaposing the histories of artemisinin research and *changshan* research from the 1940s. In contrast to this early period of experimentation, it became both possible and scientifically legitimate to research Chinese drugs in ways that had previously provoked controversy and conflict. Moreover, because Maoist China conceptualized science as constituted by both *tu* and *yang* dimensions, modern medical research could harness the productive energies of mass participation to technical expertise in and outside the laboratory. The mutual intertwining of *tu* and *yang* dimensions of modern Chinese science generated powerful results in the case of artemisinin.

Conclusion

The discovery of artemisinin is a tale whose most captivating dimensions remain invisible if we assume the stability of scientific values across both time and space. It is a story that has taken twenty some years to craft, and in its most popular form blends the achievements of an individual with the advancements of society. When taken alone, it is easy to accept this narrative of discovery as an earnest celebration of the power of science to overcome adversity. But if re-contextualized, the story of artemisinin highlights the significance of the sociopolitical context for the production of knowledge in socialist-era China. *Tu* science was not a slogan. It infused and structured social relationships as well as recategorized what sorts of activities and resources (intellectual, natural, and material) constituted scientific endeavor. The complex struggle of epistemic legitimation for Chinese drugs did not end during the Maoist period, but under the banner of *tu* science, Chinese drugs like *qinghao* came to embody the untapped potential of native and indigenous practices. Through its emphasis on mass mobilization and the integration of research and extension, *tu* science transformed popular use of *qinghao* into valuable, viable, and scientifically recognizable clinical evidence upon which further investigations—extraction, animal experiment, clinical trials, synthesis—could advance. Within this setting, *tu* science enabled Project 523 to utilize a broad range of personnel and practices in its pursuit of new antimalarial drugs.

Understanding how *tu* science shaped the sociopolitical context of knowledge production in socialist China gives us a way to think through other similar scientific endeavors, such as the discovery of arsenic trioxide for the treatment of acute promyelocytic leukemia and the synthesis of crystalline insulin—two other notable accomplishments

³⁷ For more on the struggle over Chinese drugs during the 1940, see Lei, *Neither Donkey nor Horse* (ref. 8), esp. ch. 9.

旨在快速工业化和集体化的社会经济运动。³⁶毛泽东对一种既不贬低也不否定群众参与和中国/本土知识的科学的渴望，也框定了文化大革命这一高度政治化时期医学研究的开展方式。

“土”与医学（“医”）结合时，会产生微妙而不同的含义。在医学研究领域，“土”的含义多种多样，从民间和迷信到地方性（而非全国性）和本土性。这些表述都包含在对“土”的更广泛的社会和政治特权中，因为“土”唤起了自力更生、群众参与和实践经验。因此，“土”也可以指向医学科学通常不承认的经验形式。奇妙的是，青蒿素的发现说明了“土科学”如何使重新分配和重新连接中国药物的认识论价值的过程合法化，而这些药物在20世纪40年代曾备受争议。“土科学”的概念允许以一种不损害结果科学性并阻止青蒿素作为最有效的抗疟药物之一获得国际认可的方式重新安排研究方案。

青蒿素的案例为我们提供了一个机会，来思考文化大革命时期的中国现代科学是如何在“土科学”的旗帜下运作的。这并不是说毛泽东时代的医学研究很少或根本没有“洋”的特征；也不应得出结论，认为强调“土科学”必然意味着拒绝“洋科学”，尽管两者之间存在张力。篇幅所限，无法更全面地探讨医学研究的“洋”方面，但足以说明研究人员所期望和展现出的高水平技术专长、持续创建用于高级培训和教学的专业机构、民用和军事研究机构的组织结构以及所使用的通用研究方案和设备，都表明了“洋科学”的存在和重要性。本文的兴趣在于描绘“土”对“洋”的影响。换句话说，对以“523项目”为代表的“土科学”的探索表明，毛泽东时代中国对何为科学的不同评价，产生了在其他情况下会被排除的认识论和方法论可能性，特别是当我们比较“523项目”的科学实验与国民党对常山的研究所时。

因为“土”和“洋”被认为是科学研究的构成要素，所以我认为在文化大革命时期，可能性和科学合法性的空间运作方式有所不同。在“土科学”的指导下，中药，即传统药材，

³⁶ 例如，参见陈志言，“不信土法就是不信群众” [“Disbelief in native methods is the same as disbelief in the masses”], 《人民日报》（1958年7月8日）；“土专家和洋专家的经验汇流” [“The coming together of local and foreign experts”], 《人民日报》（1958年7月14日）；金文，“走群众路线普及科学研究” [“Taking the mass line to popularize scientific research”], 《人民日报》（1958年7月20日）。

民间疗法和秘方，以及与其治疗功效相关的经验知识，都得到了提升和重视。尽管毛泽东时代对中药的兴趣本身并非新鲜事，但在“土科学”的旗帜下，将中药的认识论价值与传统医学的其他要素分离和划界的激烈斗争在很大程度上被消解了。³⁷通过将青蒿素研究和20世纪40年代常山研究的历史并置，我们可以最清楚地看到这一点。与早期的实验时期相比，以过去曾引发争议和冲突的方式研究中药变得既可能又具有科学合法性。此外，由于毛泽东时代的中国将科学概念化为由“土”和“洋”两个维度构成，现代医学研究可以利用群众参与的生产力，结合实验室内外技术专长。现代中国科学的“土”和“洋”维度的相互交织在青蒿素的案例中产生了强大的成果。

结论
青蒿素的发现是一个引人入胜的故事，但如果我们假设科学价值在时间和空间上都是稳定的，那么它最吸引人的维度就会变得不可见。这是一个耗时二十多年才得以形成的故事，其最流行的形式将个人成就与社会进步融为一体。单独来看，很容易将这种发现叙事视为对科学克服逆境力量的真诚颂扬。但如果重新审视其背景，青蒿素的故事则突显了社会政治背景对于社会主义时期中国知识生产的重要性。“土科学”并非一句口号。它渗透并构建了社会关系，并重新定义了哪些活动和资源（智力、自然和物质）构成了科学努力。中药的认知合法性复杂斗争并未在毛泽东时代结束，但在“土科学”的旗帜下，青蒿等中药开始体现出本土和传统实践的未开发潜力。通过强调群众动员以及研究与推广的结合，“土科学”将青蒿的普及使用转化为有价值、可行且科学上可识别的临床证据，在此基础上，进一步的调查——提取、动物实验、临床试验、合成——得以推进。在这种背景下，“土科学”使523项目能够利用广泛的人员和实践来寻求新的抗疟药物。

理解“土科学”如何塑造了社会主义中国知识生产的社会政治背景，为我们思考其他类似科学探索提供了途径，例如发现三氧化二砷治疗急性早幼粒细胞白血病以及合成结晶胰岛素——这两项都是显著的成就。

³⁷ 关于20世纪40年代中药之争的更多信息，请参见Lei的《非驴非马》（参考文献8），特别是第9章。

of Mao-era socialist science.³⁸ In the case of arsenic trioxide, we see similar points of how and why the Health Bureau of Heilongjiang province came to ask Zhang Tingdong, a physician who had been trained in biomedicine at Harbin Medical University and in the 1960s took classes in traditional Chinese medicine, to examine the validity of

efficacy claims made by a countryside practitioner found to be injecting a combination of arsenic, mercury, and toad venom to treat lymphatic tuberculosis and cancers. If the story of the discovery of artemisinin provokes slight discomfort in contemporary China, it may be because socialist valuations of science have not been entirely overridden.

³⁸ See Rao Yi, Li Runhong, and Zhang Daqing, “A Drug from Poison: How the Therapeutic Effect of Arsenic Trioxide on Acute Promyelocytic Leukemia Was Discovered,” *Science China* 56, no. 6 (June 2013): 495–502, and Xiong Weimin and Wang Keding, *Hecheng yige danbaizhi: jiejing niu yidaosu de rengong quanhecheng* [Synthesize a protein: The story of total synthesis of crystalline insulin project in China] (Jinan: Shandong jiaoyu chubanshe, 2005).

毛泽东时代社会主义科学的体现。³⁸ 就三氧化二砷而言，我们看到了类似的观点，即黑龙江省卫生局是如何以及为何要求张亭栋——一位曾在哈尔滨医科大学接受生物医学培训，并在20世纪60年代学习过中医的医生——来检验一位乡村医生所提出的疗效声明的有效性。

这位乡村医生被发现注射砷、汞和蟾蜍毒液的混合物来治疗淋巴结核和癌症。如果青蒿素的发现故事在当代中国引起了些许不安，那可能是因为社会主义的科学价值观尚未完全被取代。

³⁸ 参见饶毅、李润宏和张大庆，“A Drug from Poison: How the Therapeutic Effect of Arsenic Trioxide on Acute Promyelocytic Leukemia Was Discovered”，《中国科学》56卷第6期（2013年6月）：495-502页；以及熊卫民和王克定，《合成一个蛋白质：结晶牛胰岛素的人工全合成 [合成一个蛋白质：中国结晶胰岛素全合成项目的故事]》（济南：山东教育出版社，2005年）。